CYBERSECURITY / TECHNICAL PORTFOLIO

Fairway Accounting
Group Security Overhaul

Security assessment, remediation, and validation for a fictional 45-employee accounting firm.

ASSESS → IMPROVE → TEST → DOCUMENT

PROJECT AT A GLANCE

- Environment: VirtualBox lab with pfSense, Windows Server, Windows 11, Ubuntu, Kali Linux, and Wazuh.
- Focus: network access, Active Directory permissions, authentication hardening, HTTPS, SMB signing, and monitoring.
- Evidence: before/after configuration checks, Nmap scans, access tests, and Wazuh failed-login alerts.
- Scope: representative systems and controlled tests, not a production rollout to 45 employees.

PORTFOLIO PURPOSE

A concise record of implemented controls, measured results, technical troubleshooting, and remaining security risks.

Architecture & Lab Scope

Representative network topology and targeted web-server isolation

The initial environment placed important systems on a broadly accessible internal network. The lab introduced a separate 10.10.20.0/24 web segment through pfSense OPT1 while retaining the 10.10.10.0/24 internal LAN.

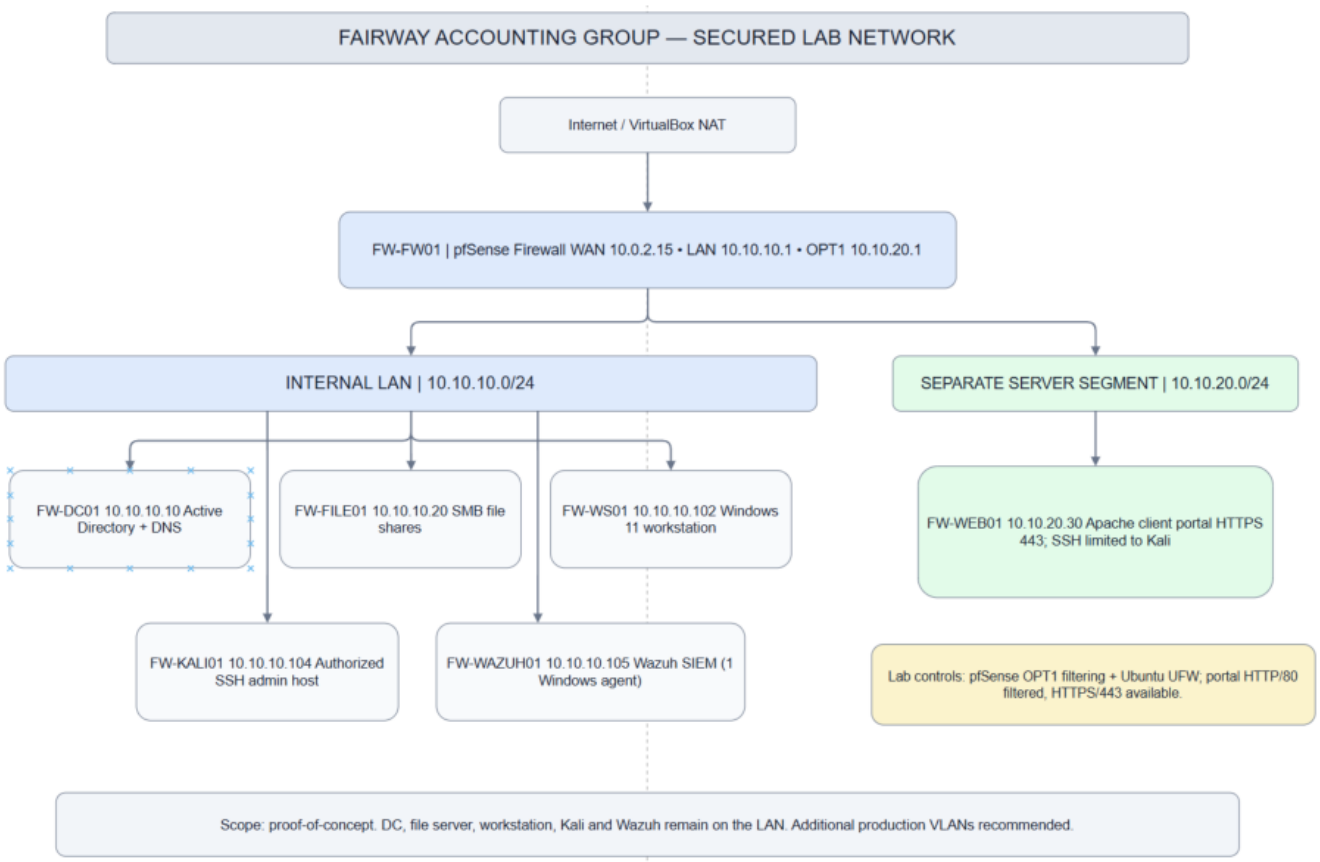


Figure 1 • Secured lab network architecture from the final audit report

- Internal: Active Directory/DNS, file server, Windows workstation, Kali testing host, and Wazuh.
- Web segment: HTTPS client portal; access restrictions tested from authorized and unauthorized hosts.
- Limitation: internal LAN rules remained broad; this was partial segmentation, not a full VLAN deployment.

Risk Findings & Remediation

Selected findings and verified changes

Excessive Payroll access

Incorrect AD group membership removed; access retest denied.

Weak domain password policy

Minimum increased from 7 to 12 characters; lockout after 5 failures.

Unencrypted web portal

HTTPS enabled; direct HTTP filtered in follow-up scan.

SMB signing optional

File server configured to require SMB signing.

Limited monitoring

Wazuh installed; one Windows workstation enrolled for log collection.

RISK ASSESSMENT

Eight risks assessed in the original report. Residual risk remained after selected fixes because the lab was not a full production implementation.

Verification Evidence

Network scanning and access-control validation

WEB PORTAL SERVICE SCAN

```
┌──(securityadmin㉿fw-kali01)-[~]
└─$ sudo nmap -Pn -sV -p 22,80,443 10.10.20.30
[sudo] password for securityadmin:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-06 21:27 -0400
Nmap scan report for 10.10.20.30
Host is up (0.0011s latency).

PORT    STATE    SERVICE  VERSION
22/tcp  open     ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.19 (Ubuntu Linux; protocol 2.0)
80/tcp  filtered http
443/tcp open     ssl/http Apache httpd 2.4.58 ((Ubuntu))
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.03 seconds
```

Figure 2 • Nmap: HTTPS 443 open, HTTP 80 filtered, SSH 22 open from Kali

FILE SHARING & LEAST PRIVILEGE

```
┌──(securityadmin㉿fw-kali01)-[~]
└─$ sudo nmap -Pn -p 445 --script smb2-security-mode 10.10.10.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-06 21:29 -0400
Nmap scan report for 10.10.10.20
Host is up (0.00075s latency).

PORT    STATE SERVICE
445/tcp open  microsoft-ds
MAC Address: 08:00:27:33:30:A4 (Oracle VirtualBox virtual NIC)

Host script results:
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled and required

Nmap done: 1 IP address (1 host up) scanned in 1.67 seconds
```

SMB security-mode validation confirmed signing was enabled and required for SMB 3.1.1.

An access retest also confirmed the Accounting user could no longer open the Payroll share after incorrect group membership was removed.

Security Monitoring & Detection

Controlled authentication-failure simulation

A Windows 11 workstation was enrolled as a Wazuh agent. Three intentional failed logins were generated using a lab account, below the configured five-attempt lockout threshold.

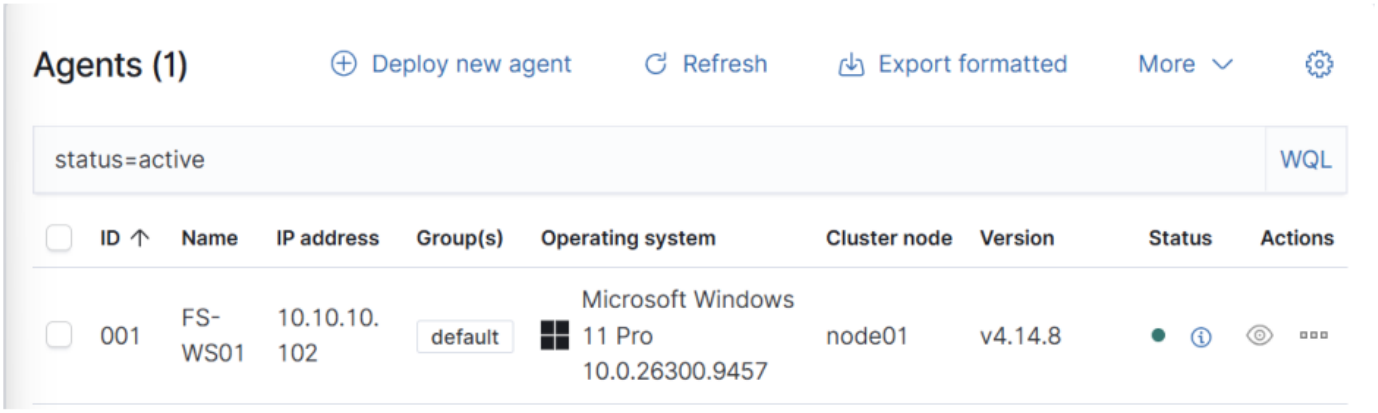


Figure 3 • Wazuh agent status: Windows workstation reporting to manager

WINDOWS EVENT

Three failed sign-in events were recorded as Windows Event ID 4625.

WAZUH RESULT

Three Level 5 alerts matched Wazuh rule 60122, confirming event collection and detection.

This was a controlled detection test, not a confirmed breach or full incident-response exercise.

Troubleshooting & Technical Judgment

Root-cause investigation and documented limitations

DNS forwarding / Wazuh installation

The Windows workstation could not resolve a Wazuh download hostname. Testing traced the failure to domain-controller DNS forwarding; external resolution succeeded after the DNS forwarder was corrected.

Network segmentation / firewall

Moving the portal required adapter, subnet, gateway, pfSense and Ubuntu firewall changes. HTTPS and SSH tests from different hosts were used to verify the intended restrictions.

Wazuh service health

Investigated manager, indexer and dashboard services and available VM storage. Expanded storage during setup and restarted the affected indexer service to restore dashboard availability.

Ubuntu VM startup

Some Ubuntu VMs required GRUB boot-parameter workarounds, including nomodeset. This was not permanently resolved and remained a lab limitation.

Results, Limits & Next Steps

Evidence-based outcomes and a realistic roadmap

- Restricted unauthorized Payroll share access and verified the denied result.
- Strengthened Active Directory password and account-lockout requirements.
- Enabled HTTPS, required SMB signing, and tested exposed network services.
- Confirmed Wazuh collected and displayed three simulated authentication failures.
- Produced an audit report, risk register, incident response plan, architecture diagrams, and troubleshooting record.

REMAINING WORK FOR A REAL DEPLOYMENT

- Expand internal VLAN segmentation and reduce broad LAN access.
- Deploy MFA, trusted production certificates, and broader endpoint monitoring.
- Test backup restoration and exercise the proposed incident-response plan.
- Assess production risks with business stakeholders, including documented risk acceptance decisions.

SOURCE MATERIAL

Compiled from the five original Fairway capstone PDFs: security audit, presentation, troubleshooting, incident response plan, and cloud/security architecture. Proposed cloud controls were not deployed.